

JUNQI LU

[GitHub](#) — [Google Scholar](#) — [Personal Website](#)

Date of Birth: August 24, 2002 — Email: lu.junqi.001@student.uni.lu

Luxembourg

EDUCATION

University of Luxembourg

Belval, Luxembourg

- **M.S. in Mathematics**

Sep. 2026 – Jun. 2028 (Expected)

Beijing Institute of Technology

Beijing, China

- **B.E. in Computer Science** (Second Bachelor's Degree)

Sep. 2024 – Jun. 2026

- **B.S. in Mathematics and Applied Mathematics**

Oct. 2020 – Jun. 2024

PUBLICATIONS

(*: Equal contribution; #: Corresponding author)

[1] **Junqi Lu***, Ruixiang Sun*, Xin Li[#], Gaopeng Peng, Ao Zhang, and Mingzhong Wang[#]. Prevailing Bisimulation Metric Learning Is Biased: Implicit Regularization and Its Remedy. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. (Accepted)

[2] **Junqi Lu**, Bosen Liu, Cuicui Pei, Qingan Qiu[#], and Li Yang[#]. Learning to optimize termination decisions under hybrid uncertainty of system lifetime and task duration. *Computers & Industrial Engineering*, 206, 111208, 2025. DOI: [10.1016/j.cie.2025.111208](https://doi.org/10.1016/j.cie.2025.111208) (Published)

[3] **Junqi Lu**, Qingan Qiu[#], Rongchi Sun, and Yike Zhang. Simplicity Wins: How Myopic Decisions Enable Safe Learning in Mission-Critical Systems. (Under Review)

[4] Rongchi Sun, Yike Zhang, **Junqi Lu**, and Qingan Qiu[#]. Dynamic Switching Strategy for Standby Systems with Switching Time Redundancy. (Under Review)

RESEARCH EXPERIENCE

Deep Reinforcement Learning & State Representation

Sep. 2025 – Jun. 2026

Research Assistant | Advisor: [Prof. Xin Li](#) | [1029 DRL Lab](#) | Beijing Institute of Technology

- **Theory & Methodology**: Identified the metric learning "Variance Trap" via **SDEs** and proposed **BSPO** (*Bisimulation Saddle-Point Optimization*), a primal-dual framework utilizing an auxiliary network to decouple structural error from transition noise.
- **Evaluation**: Evaluated BSPO on **DeepMind Control Suite** under stochastic dynamics and visual distractions, and on **Atari**, comparing against MICO and SimSR baselines.
- **Outcome**: Led to a first-author paper accepted at **NeurIPS 2026**.

Reliability Engineering & Stochastic Control

Jun. 2023 – Jun. 2026

Research Assistant | Advisor: [Prof. Qingan Qiu](#) | Beijing Institute of Technology

- **Bayesian MDP Framework**: Formulated dynamic condition-based task termination under hybrid uncertainty (unknown system lifetime and task duration distributions) within an MDP framework, establishing conjugate Gamma-exponential Bayesian parameter updating with censored execution data.
- **Theoretical Insights & Algorithm**: Proved the existence of optimal termination thresholds, structural monotonicity of value functions, and monotonic threshold dynamics; designed an $\mathcal{O}(N)$ monotone iterative learning algorithm that cuts operational and failure costs by up to 45.11%.
- **Subsequent Extensions & Publications**: Extended foundational framework to myopic safe-learning heuristics and dynamic switching under redundancy. Published 1 first-author paper in **CAIE** (JCR Q1, IF=6.5); 2 first-/co-authored papers under review.

SELECTED PROJECTS

- Nitrogen Cycling Dynamics Model (MCM-ICM 2025 Finalist)

[GitHub: 100+ Stars]
- **Modeling:** Developed a dynamic ecosystem model using modified **Lotka-Volterra equations** to simulate nitrogen flow across forest and agricultural food webs.
 - **Open Science:** Released a fully reproducible solution repository with comprehensive documentation and tutorials. Garnered **100+ GitHub Stars**, widely recognized by the modeling community for its code quality and educational value.
 - **Award:** Selected as a **Finalist Winner (Top 2%)** out of 27,456 teams globally.

- Formal Verification of a Mahjong Winning-Hand Checker

[Demo] [Code]

AI4Math Summer School, Zhejiang University

Summer 2026
- **Independent Project:** Conceived and completed the project individually, from problem selection and specification design to implementation, formal proofs, and the final summer-school presentation.
 - **Formal Modeling:** Encoded three-suit Mahjong in **Lean 4**, separating declarative winning-hand specifications from an executable checker for standard hands and seven pairs, including declared melds and kongs.
 - **Correctness Proofs:** Proved the executable checker sound and complete with respect to the declarative specification for every hand state: `isWinning state = true` $\leftrightarrow$ `WinningSpec state`, extending validation beyond individual test cases.

HONORS & AWARDS

- **Meritorious Winner**, Mathematical Contest in Modeling (MCM) 2023, 2026
- **Outstanding Student Award** (Top 10% university-wide), BIT Nov. 2025
- **Academic Excellence Scholarship** (Rank 1st in CS Program), BIT Oct. 2025
- **Finalist (Top 2%)**, Interdisciplinary Contest in Modeling (ICM) Feb. 2025
- **University “Excellent Thesis” Prize**, BIT Jun. 2024
- **Second Prize**, China Undergraduate Mathematical Contest in Modeling (CUMCM) Nov. 2022

SKILLS

Programming Languages	Python (Expert in PyTorch, NumPy, Pandas), C++ (Linux System Dev), SQL, MATLAB, Bash
AI Workflow	Chinese (Native), English (B2), German (A1)
Tools	Proficient in AI-assisted coding (VS Code + Copilot/Cursor) for efficient development.
Outreach	Git, Docker, LaTeX, Makefiles, Lean 4 (Formal Proof Verification)
	Tech Blogger (CNBlogs) with 20K+ views (20+ articles on DSA & LeetCode).

INTERESTS

- **Marathon Running:** Committed long-distance runner with an annual mileage exceeding **1,000 km** for three consecutive years. Half-Marathon Personal Best: **1:41:34**.
- **Music:** Lead guitarist and bassist for university rock bands. Passionate about classic rock arrangement and performance; active participant in campus music festivals.